
RP-GAAP: Global Attribute-Association Pattern Aggregation with Rare-Pattern Weighting for Graph Fraud Detection

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Abstract

GAAP achieves state-of-the-art graph fraud detection by learning attribute-association patterns through adaptive feature binning and message passing. However, its standard cross-entropy loss treats every training node equally, causing rare fraud-indicative patterns to be drowned out by common benign ones. We propose RP-GAAP, which assigns higher sample weights to nodes with infrequent patterns, improving detection without any architectural modification. Preliminary results show Recall@K gains of +0.97 on T-Finance and +0.78 on Tolokers. Project page: <https://github.com/NathanTrance/GraphML>.

1 Introduction

Fraud on online platforms—fake reviews, money laundering, collusive crowdsourcing—causes billions in annual losses. Detecting these activities is inherently a graph problem: fraudsters interact with victims, form collusive rings, and leave topological traces invisible to non-relational models [12, 1]. Graph Neural Networks (GNNs) have become the dominant approach because they jointly model node attributes and neighborhood structure [4, 3].

However, fraud detection presents two hard challenges for standard GNNs. First, camouflage: fraudsters mimic benign feature distributions to blend in [1]. Second, neighborhood inconsistency: a fraudster’s neighbors are mostly benign, diluting the fraud signal during message passing [7]. On top of these, class imbalance is severe—fraud typically constitutes 3–17% of nodes, so a model predicting “benign” for all achieves high accuracy while catching zero fraud. Existing detectors address these via neighbor filtering or sampling [1, 6, 7], adding complexity without treating pattern rarity itself. Focal loss [5] reweights by classification difficulty, not intrinsic rarity.

GAAP [2] departs from neighbor engineering by learning *attribute-association patterns*: adaptive feature binning preserves per-attribute semantics, message passing extracts neighborhood association context, and a global cross-attention module captures long-range correlations. GAAP achieves SOTA against 24 baselines on 7 datasets without neighbor filtering. However, it trains with standard cross-entropy, assigning equal weight to every node. Common benign patterns dominate the gradient, while rare fraud-indicative patterns receive negligible attention. Our central question: can we improve GAAP simply by telling the optimizer which training examples encode rare patterns?

We propose RP-GAAP (Rare-Pattern-weighted GAAP), a lightweight extension that computes per-sample weights from quantile-binned feature pattern frequencies and uses these scores in the cross-entropy loss. The method adds three hyperparameters, introduces no learned parameters, and requires no architectural changes. We evaluate RP-GAAP on four fraud detection benchmarks (YelpChi, T-Finance, Elliptic, Tolokers) across four loss configurations.

2 Proposed Method

2.1 Preliminaries

Let $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ be an attributed graph with $N = |\mathcal{V}|$ nodes. Each node i has a feature vector $\mathbf{x}_i \in \mathbb{R}^d$ and a binary label $y_i \in \{0, 1\}$ (0 = benign, 1 = fraud). The nodes are partitioned into disjoint training, validation, and test sets. Fraud typically constitutes 3–17% of nodes, making the problem severely class-imbalanced.

2.2 Background: GAAP

GAAP [2] produces a node representation $\mathbf{h}_i \in \mathbb{R}^D$ for each node i through three stages, then predicts $\hat{y}_i = \text{softmax}(\mathbf{W}\mathbf{h}_i)$. It is trained with standard cross-entropy:

$$\mathcal{L}_{\text{GAAP}} = -\frac{1}{|\mathcal{T}_{\text{train}}|} \sum_{i \in \mathcal{T}_{\text{train}}} \log \hat{y}_{i, y_i} \quad (1)$$

2.2.1 Dynamic Binning Embedding (DyBEM)

Raw continuous features cannot be fed directly into message passing without fusing distinct semantics. DyBEM addresses this by learning b adaptive bin boundaries per feature dimension. For the j -th feature, the bin boundaries $\{q_{j,0}, q_{j,1}, \dots, q_{j,b}\}$ are initialized via decision-tree binning on the training set [2] and optimized jointly with the GNN. Each scalar value $x_{i,j}$ is then represented as a b -dimensional vector where the m -th component encodes the fragment-wise membership of $x_{i,j}$ in bin m , producing a bin embedding $\mathbf{e}_{i,j} \in \mathbb{R}^b$. Features from all d dimensions are stacked to form $\mathbf{E}_i \in \mathbb{R}^{d \times b}$, which is flattened and projected to the GNN’s hidden dimension.

2.2.2 Message-Passing with GraphSAGE

GAAP employs GraphSAGE [3] with max-pooling aggregation as its GNN backbone. For each node v at layer $l + 1$, the update rule is:

$$\mathbf{h}_v^{(l+1)} = \sigma \left(\mathbf{W}_{\text{self}}^{(l)} \mathbf{h}_v^{(l)} + \mathbf{W}_{\text{neigh}}^{(l)} \cdot \max_{u \in \mathcal{N}(v)} \sigma_{\text{pool}}(\mathbf{W}_{\text{pool}}^{(l)} \mathbf{h}_u^{(l)}) \right) \quad (2)$$

where $\mathbf{h}_v^{(0)} = \text{MLP}(\mathbf{E}_v)$ is the DyBEM-projected feature, $\mathcal{N}(v)$ denotes the neighborhood of v , and σ is a ReLU activation. Max-pooling is chosen because it preserves the strongest signals per dimension rather than averaging them away—critical when a fraud node is surrounded by mostly benign neighbors. The output of the final GNN layer is denoted $\mathbf{g}_i \in \mathbb{R}^{d_h}$, representing the node’s local *association pattern*.

2.2.3 Global Pattern Aggregation

To mitigate over-smoothing and capture long-range dependencies across the graph, GAAP applies a multi-head cross-attention module. Rather than computing pairwise attention across all N nodes (prohibitively expensive), GAAP caches the GNN output from the previous epoch as historical embeddings $\tilde{\mathbf{G}} \in \mathbb{R}^{N \times d_h}$. For each node i , the query is its current GNN output and the keys/values are the cached embeddings:

$$\mathbf{o}_i = \text{MHA}(\mathbf{q}_i = \mathbf{g}_i, \mathbf{K} = \tilde{\mathbf{G}}, \mathbf{V} = \tilde{\mathbf{G}}) \quad (3)$$

The final representation is a residual combination $\mathbf{h}_i = \alpha \cdot \mathbf{o}_i + (1 - \alpha) \cdot \mathbf{W}_{\text{out}} \mathbf{g}_i$, where α is a learned or fixed mixing coefficient.

2.3 RP-GAAP: Rare-Pattern Weighted Training

Under GAAP’s equal-weighted loss (Eq. 1), all training nodes contribute identically to the gradient. Yet the distribution of attribute-association patterns is highly skewed: common patterns dominate,

while rare patterns—disproportionately associated with fraud—receive negligible gradient signal. RP-GAAP addresses this by precomputing a rarity weight w_i for each training node.

Pattern construction. We select the top- k features by variance over $\mathcal{T}_{\text{train}}$. For each selected feature j , we partition its training-set values into b quantile bins with boundaries $q_{j,0} < \dots < q_{j,b}$. Node i is assigned bin index $c_{i,j}$ where $q_{j,c_{i,j}} \leq x_{i,j} < q_{j,c_{i,j}+1}$. The k bin indices are combined into a single pattern identifier:

$$p_i = \sum_{j \in \mathcal{S}} c_{i,j} \cdot b^{\pi(j)} \quad (4)$$

where $\mathcal{S}$ is the set of selected feature indices and $\pi(j)$ maps each feature to its position in the set.

Weight assignment. Let $f(p) = |\{i \in \mathcal{T}_{\text{train}} : p_i = p\}|$ be the frequency of pattern p in the training set. We linearly interpolate between 1.0 (most frequent) and w_{max} (rarest), then apply a multiplicative boost $\phi \geq 1$ to fraud nodes, clipped to $[1.0, w_{\text{max}} \cdot \phi]$:

$$w_i = \min\left(\left[1.0 + (w_{\text{max}} - 1.0) \cdot \frac{\max_q f(q) - f(p_i)}{\max_q f(q) - \min_q f(q)}\right] \cdot \phi^{\mathbb{I}_{[y_i=1]}}, w_{\text{max}} \cdot \phi\right) \quad (5)$$

Nodes not in $\mathcal{T}_{\text{train}}$ receive $w_i = 1.0$. The weight vector $\mathbf{w} \in \mathbb{R}^N$ is computed once before training on CPU (adding < 1 second overhead) and requires no gradients.

RP-GAAP loss. We replace the uniform average in Eq. 1 with a weighted sum:

$$\mathcal{L}_{\text{RP-GAAP}} = - \frac{1}{\sum_{i \in \mathcal{T}_{\text{train}}} w_i} \sum_{i \in \mathcal{T}_{\text{train}}} w_i \cdot \log \hat{y}_{i,y_i} \quad (6)$$

The GAAP forward pass (DyBEM $\rightarrow$ GraphSAGE $\rightarrow$ MHA $\rightarrow$ classifier) is entirely unchanged. The method adds exactly 4 hyperparameters— $b, k, w_{\text{max}}, \phi$ —and no learned parameters. We use $b = 5, k = 10, w_{\text{max}} = 3.0, \phi = 1.5$ across all experiments.

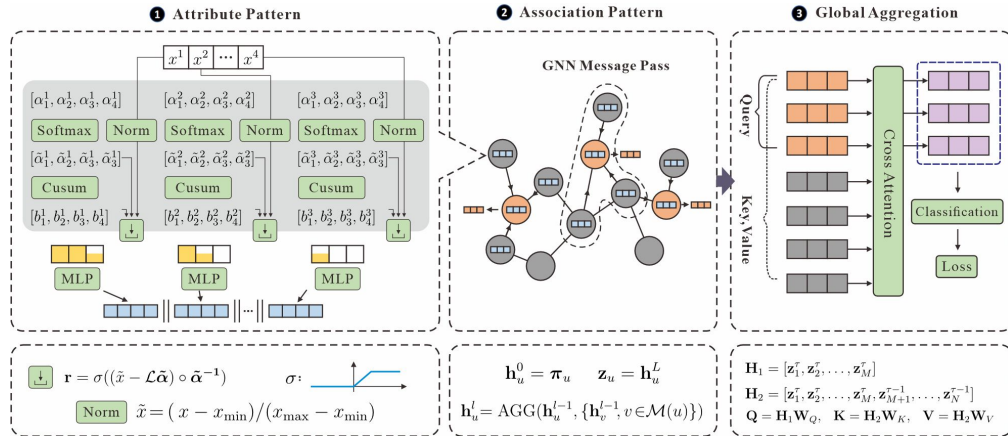


Figure 1: The GAAP framework. DyBEM creates binned attribute patterns, GraphSAGE (Eq. 2) extracts association patterns, and global cross-attention (Eq. 3) captures pattern correlations. RP-GAAP adds rarity weights (Eq. 5) to the loss without modifying this architecture. Figure adapted from Duan et al. [2].

3 Experimental Setup & Preliminary Results

3.1 Datasets and Metrics

We evaluate on four GADBench [11] benchmarks following GAAP’s splits [2]: **YelpChi** [9] (review spam, 14.5% fraud), **T-Finance** [10] (financial transactions, 3.4% fraud), **Elliptic** [13] (illicit Bitcoin, 9.5% fraud), and **Tolokers** [8] (crowdsourcing, 16.9% fraud). Following GAAP, our primary metric is Recall@K—the fraction of true fraud nodes in the top- K predictions where K equals the total number of fraud nodes; we also report AUC-ROC, AP, Macro-F1, and fraud-class Recall and Precision.

3.2 Preliminary Results

Table 1 reports Recall@K with default hyperparameters across all four loss configurations.

Table 1: Recall@K on four datasets. Best per dataset in bold.

Dataset	GAAP	+Focal	RP-GAAP	+Both	Best Δ
YelpChi	0.8839	0.8823	0.8862	0.8700	+0.23
T-Finance	0.8488	0.8530	0.8585	0.8571	+0.97
Elliptic	0.7322	0.7359	0.7313	0.7322	+0.37
Tolokers	0.5498	0.5296	0.5467	0.5576	+0.78

RP-GAAP improves over the GAAP baseline on YelpChi (+0.23) and T-Finance (+0.97), the dataset with the most extreme class imbalance. On Tolokers, combining rare weighting with focal loss yields the best Recall@K (+0.78). Elliptic shows minimal differences, likely due to its temporal structure resisting static feature binning. As next steps, we will run ablation studies and perform hyperparameter tuning to yield the final results.

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